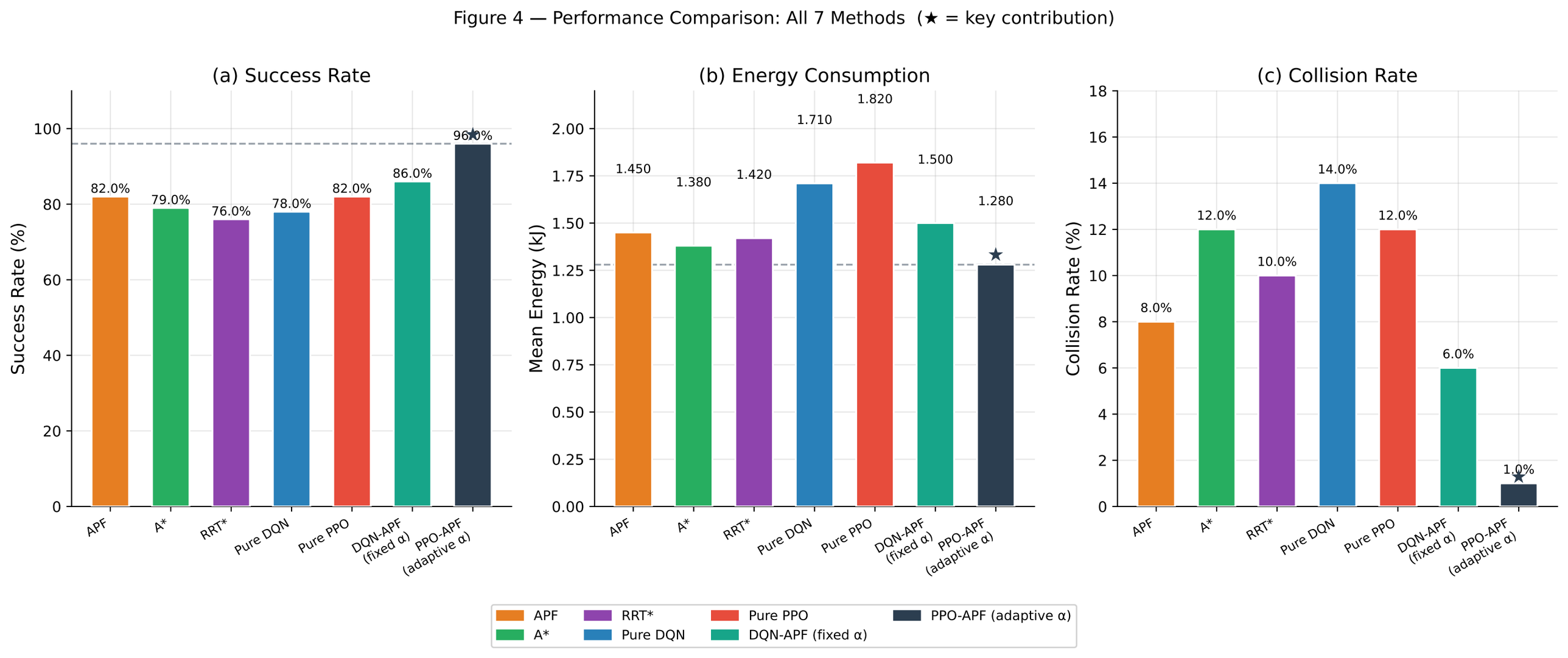
Figure 4 — Performance Comparison: All 7 Methods (★ = key contribution)
(a) Success Rate
Success Rate (%)
82.0%
79.0%
76.0%
78.0%
82.0%
86.0%
96.0%
(b) Energy Consumption
Mean Energy (kJ)
1.450
1.380
1.420
1.710
1.820
1.500
1.280
(c) Collision Rate
Collision Rate (%)
8.0%
12.0%
10.0%
14.0%
12.0%
6.0%
1.0%
APF
A*
RRT*
Pure DQN
Pure PPO
DQN-APF (fixed α)
PPO-APF (adaptive α)